Practice Quiz: Learning

Part A: Multiple Choice

1. Muscle memory in crochet is best understood in Active Inference as: A) The hands acting independently of the brain B) Precision tuning — motor predictions becoming increasingly accurate and automatic over practice C) The complete elimination of the generative model D) A permanent, unchangeable state

2. When a crocheter learns a new stitch for the first time, the initial prediction error is high because: A) The crocheter has no sensory input B) The generative model does not yet have accurate action-outcome mappings for this stitch C) The Markov blanket has dissolved D) New stitches cannot generate prediction error

3. Making a gauge swatch and adjusting hook size based on the results is an example of: A) Destroying the generative model B) Prior updating — testing a belief against evidence and revising it C) Maximizing free energy D) Random action with no learning

4. An experienced crocheter who automatically goes up one hook size because they "know they crochet tightly" is demonstrating: A) A failure of their model B) Meta-learning — a higher-level model about their own crafting tendencies C) The absence of prior beliefs D) External state control

5. A beginner's uneven stitches compared to an expert's uniform fabric reflects: A) The beginner having a better generative model B) Equal precision in both models C) The difference in motor prediction precision — the expert's model generates tighter, more accurate predictions D) The expert ignoring their model

6. Learning crochet in a circle with experienced crafters accelerates model development because: A) The beginner does not need their own generative model B) Demonstration, feedback, and shared knowledge provide richer evidence for model updating C) Social interaction eliminates prediction error D) Crochet circles prevent all mistakes

7. When a crocheter takes a long break and their first project back feels "rusty," this is because: A) Their generative model has been permanently erased B) Motor prediction precision has declined during the break, requiring re-calibration C) They have forgotten how to hold a hook entirely D) The Markov blanket no longer exists

Part B: Short Answer

1. Describe the learning process for a new stitch from first attempt to fluency. Use the concepts of prediction error, model expansion, and precision tuning.

2. Explain how prior crochet knowledge can both help and hinder learning a new related technique (such as switching from traditional crochet to Tunisian crochet).

3. A crocheter notices that their gauge changes depending on their mood — tighter when stressed, looser when relaxed. Describe this phenomenon in Active Inference terms, and suggest how awareness of this pattern could improve their crafting.